

**Question:**

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A.  $2.48 \times 10^{-8}$  cm
- B.  $5.92 \times 10^7$  cm/sec
- C.  $8.66 \times 10^{-6}$  cm
- D.  $3.14 \times 10^{-5}$  cm
- E.  $9.81 \times 10^{-4}$  cm
- F.  $5.27 \times 10^{-9}$  m
- G.  $1.23 \times 10^{-7}$  cm
- H.  $4.14 \times 10^{-10}$  m
- I.  $6.625 \times 10^{-27}$  erg sec
- J.  $0.512 \times 10^6$ eV

**Reasoning + Answer:**

<think>  
The de Broglie wavelength ( $\lambda$ ) of an electron can be calculated using the formula :  $\lambda = h / \sqrt{2 m * K}$  , where h is Planck's constant , m is the mass of the electron , and K is the kinetic energy .  
The given kinetic energy is  $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$  . Rearrange the formula to solve for  $\lambda$  :  $\lambda = h / \sqrt{2 m * K}$   
Substitute values , we get  $\lambda = h / \sqrt{2 \times 9.11 \times 10^{-31} * 1.6 \times 10^{-19}}$   
Simplify and convert to unit cm , then the de Broglie wavelength  $\lambda \approx 1.23 \times 10^{-7} \text{ cm}$  .  
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